SMART COMMERCE SYSTEM

UCE2023636 : RADHIKA KULKARNI

UCE2023637 : SWATI KUMBHAR

UCE2023646 : SHRUTIKA NALAVADE

Problem Statement ?

- Businesses struggle to make data-driven decisions.
- No unified tool for forecasting sales, predicting demand, analyzing customers, and optimizing pricing.
- Manual analysis leads to:
- Stock-outs & overstock
- Poor customer targeting
- Inaccurate forecasting
- Revenue loss

Project Overview

- Sales Forecasting
- Inventory Prediction
- Segmentation
- Association Rules
- Price Recommendation
- Analytics

Objective

- Customer segmentation
- Increase revenue
- Dynamic price recommendation
- Product demand prediction
- Personalized customer experience
- Area-wise sales analysis
- Better marketing decisions

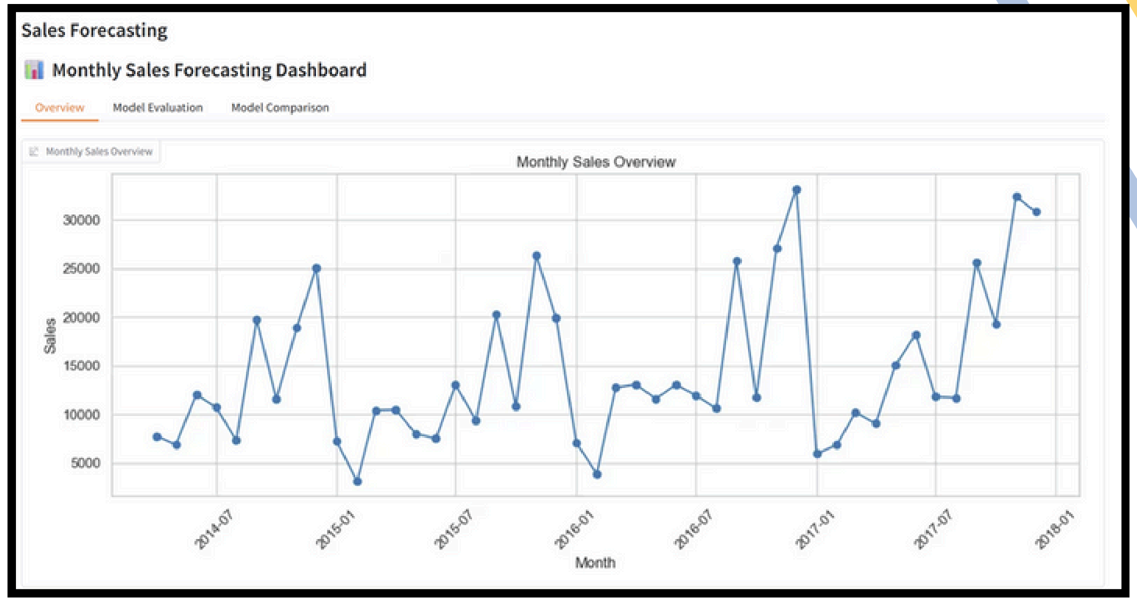
Dataset Overview

- Source: Superstore Sales Dataset
- Entries: 2,121 rows
- Features: Sales, Profit, Order Date, Customer, Region, Category, Product, etc.
- Time-series + Customer + Product features

System Architecture

- Data Preprocessing
- Sales Forecasting Module
- Inventory Prediction Module
- Customer Segmentation
- Association Rule Mining
- Price Recommendation System
- Frontend Built Using Gradio

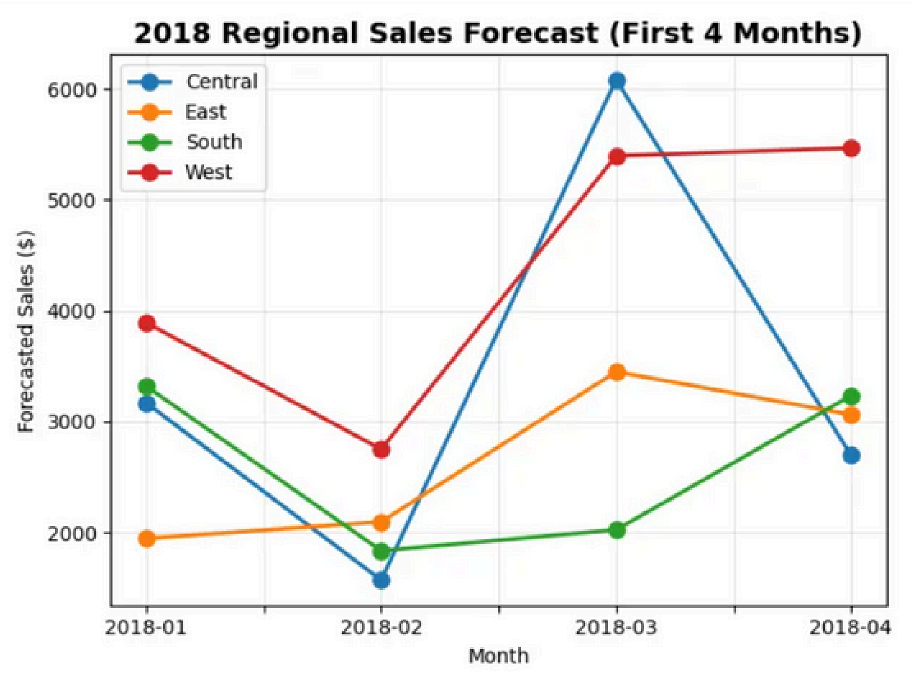
Module 1 – Sales Forecasting



- Used monthly aggregated sales with cleaned & outlier-handled data.
- Created lag + time features to capture trends and seasonality.
- Trained three models: Linear Regression, Random Forest, XGBoost.
- Linear Regression performed best (Accuracy 74.43%).
- XGBoost second (73.91%), Random Forest third (71.79%).
- Results show sales follow a strong linear trend.
- Forecasting helps in planning, budgeting, and inventory decisions.

Module 2 – Advanced Region-Based Forecasting

- Predict 2018 sales across all geographic regions
- Provide data-driven insights for strategic planning
- **Exponential Smoothing (ETS)** for time series forecasting
- Separate models for each region to capture unique patterns
- 3+ years historical sales data (2014-2017)
- Monthly aggregation by region
- Trend and seasonality analysis
- Monthly sales forecasts for Jan-Apr 2018
- Regional performance rankings
- Growth trend comparisons

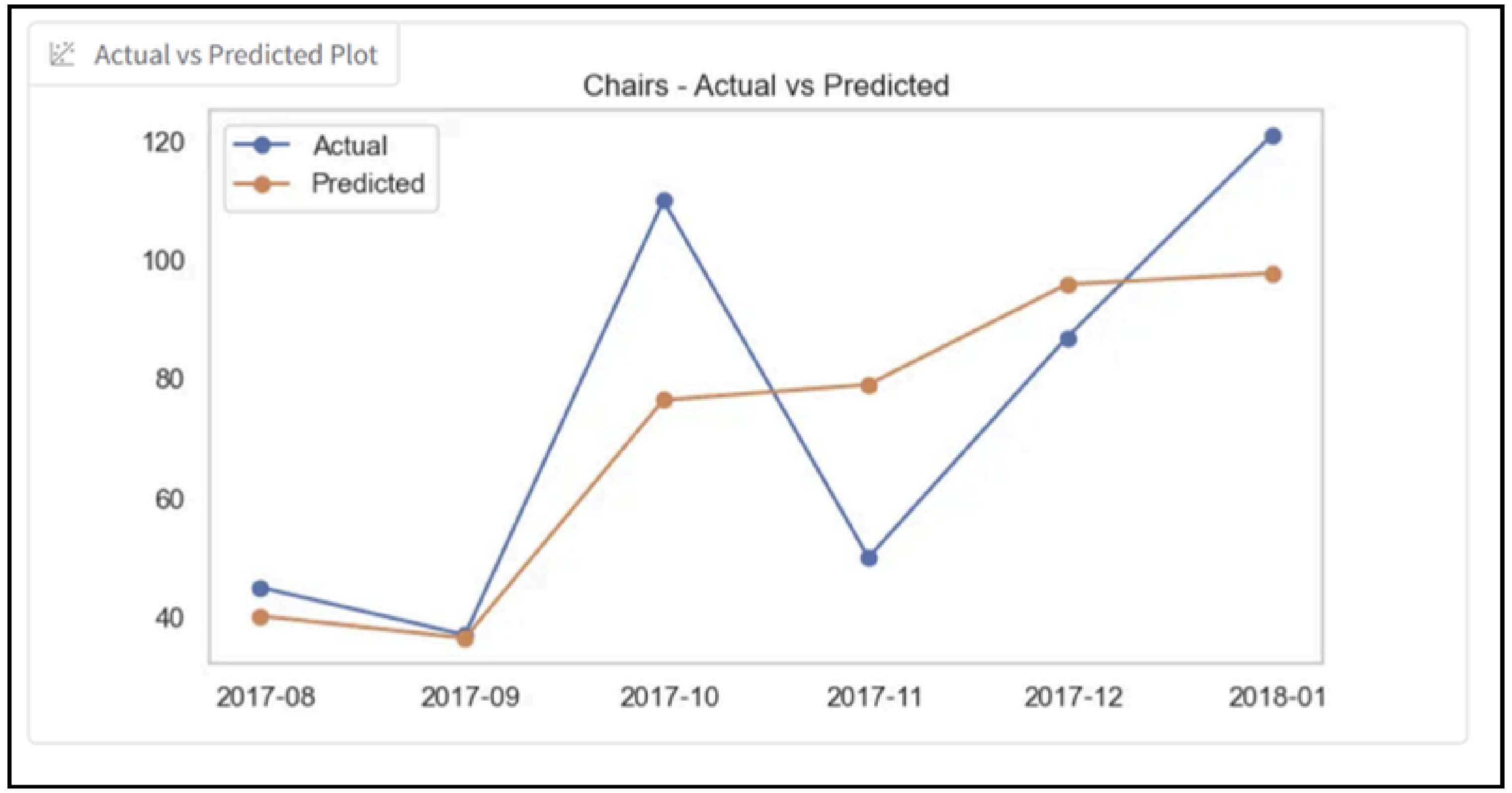


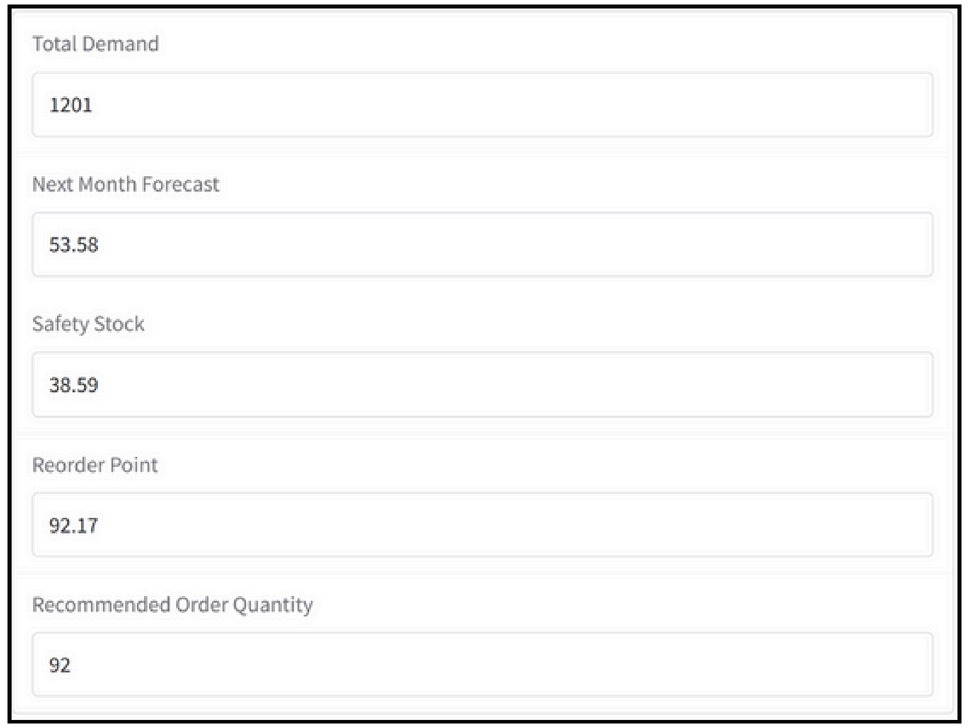
Monthly Forecast Details

Region	Month	Forecasted_Sales
Central	2018-01	3168.5891384384054
Central	2018-02	1568.754943444702
Central	2018-03	6083.084303972933
Central	2018-04	2693.0922001461176
East	2018-01	1944.2691861167405
East	2018-02	2093.951421650237
East	2018-03	3448.2912599001966
East	2018-04	3061.638734008793
South	2018-01	3318.913158958799
South	2018-02	1833.7860824966665
South	2018-03	2022.0370079222423
South	2018-04	3233.3347371823147

Module 3 – Inventory Forecasting

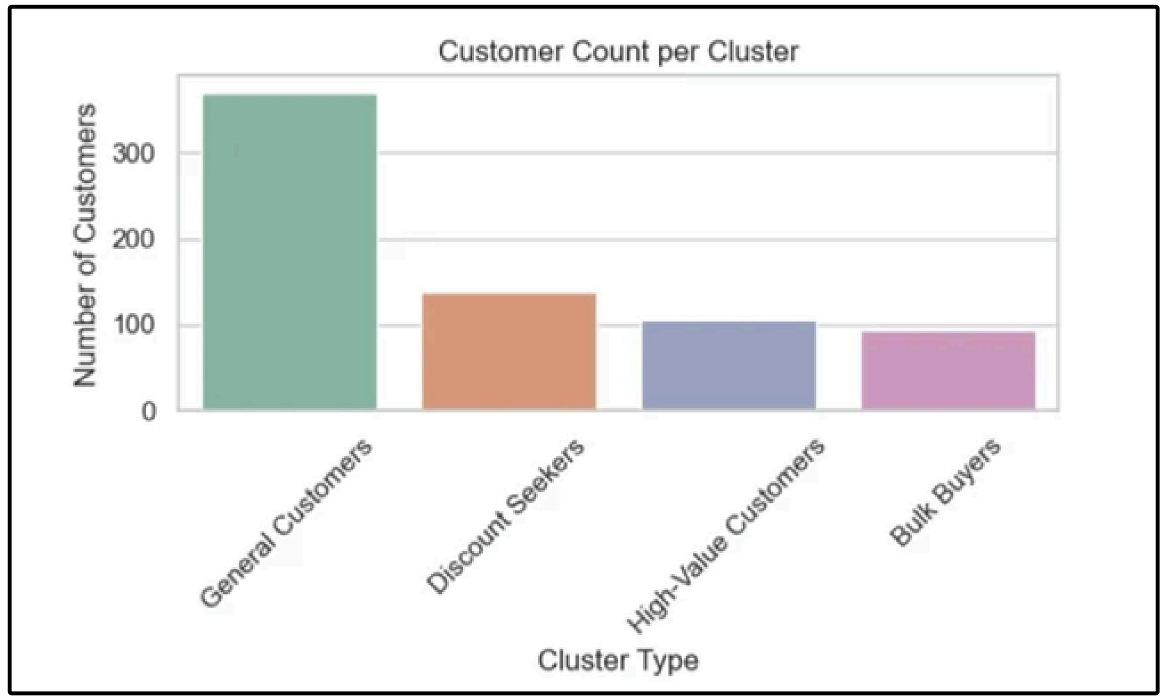
- Built a Random Forest–based demand forecasting system using monthly aggregated sales.
- Added lag features, rolling averages, and time features (Month, Year, Quarter).
- Model predicts next-month demand and evaluates performance using MAE, RMSE, sMAPE, Accuracy.
- Automatically calculates Safety Stock, Reorder Point, and Recommended Order Quantity.
- Interactive Gradio UI allows users to select any Sub-Category and view:
- Forecast metrics
- Actual vs Predicted plot
- Monthly demand table
- Inventory recommendations
- Supports real-time forecasting and helps in inventory planning & stock optimization.

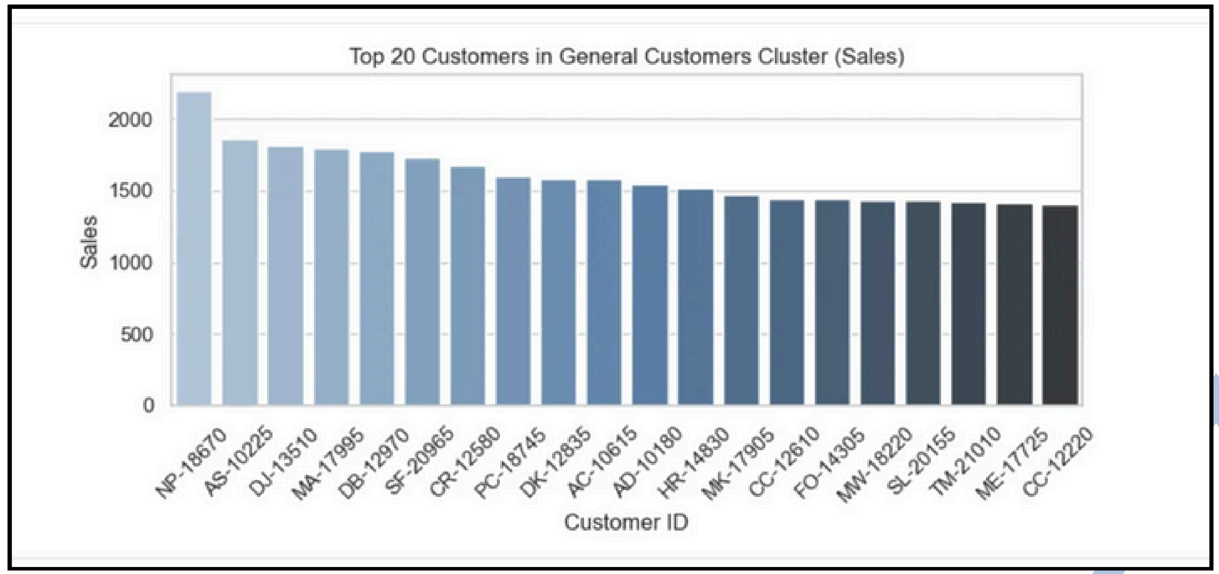




Module 4 – Customer Segmentation

- Groups customers into meaningful segments using K-Means clustering
- Uses features like Sales, Profit, Quantity, and Discount
- Identifies patterns such as High-Value, Low-Value, Bulk Buyers, Discount Seekers
- Helps understand customer buying behavior





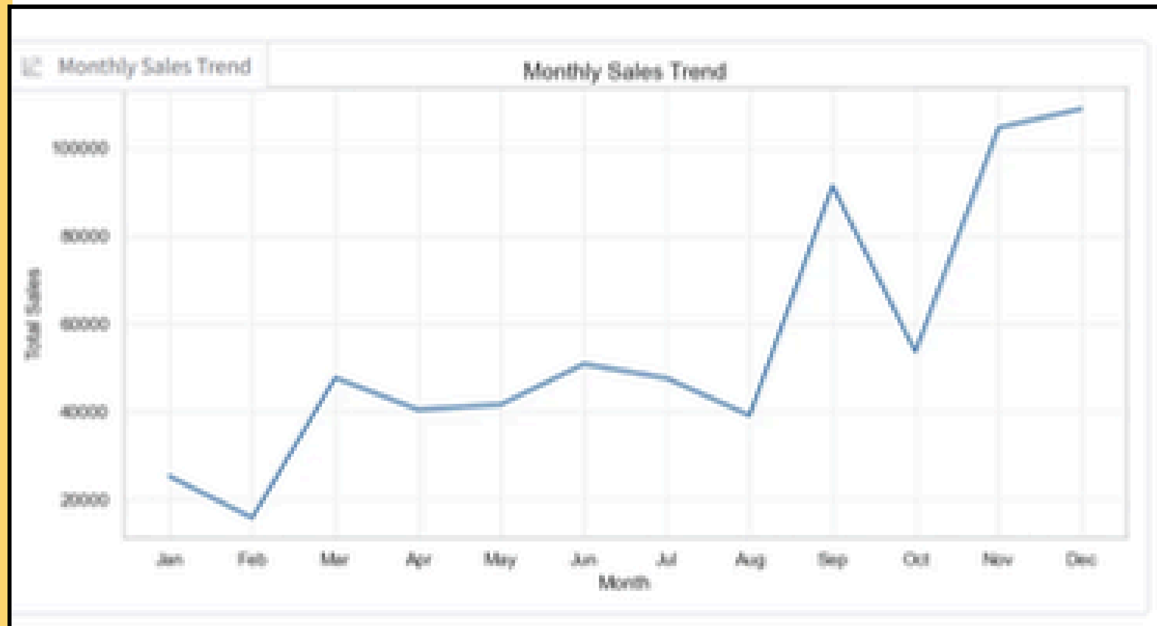
Module 5 – Association Rules

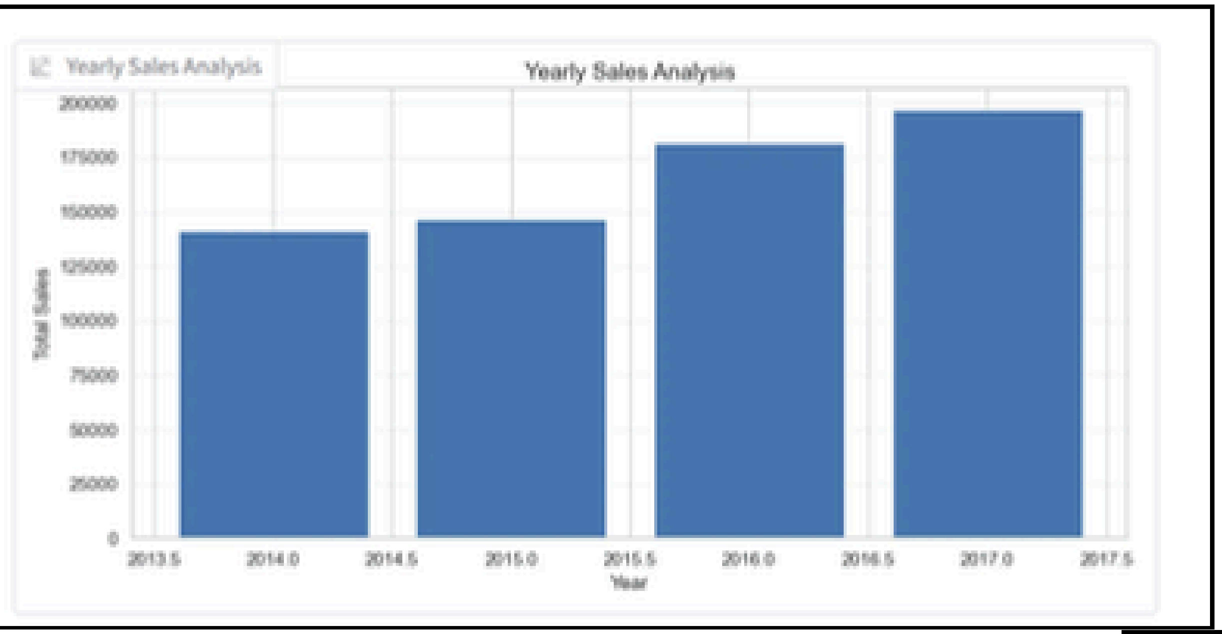
- Predicts product demand
- Suggests reorder quantity
- Helps prevent overstock & stockouts
- Uses historical sales + trend analysis

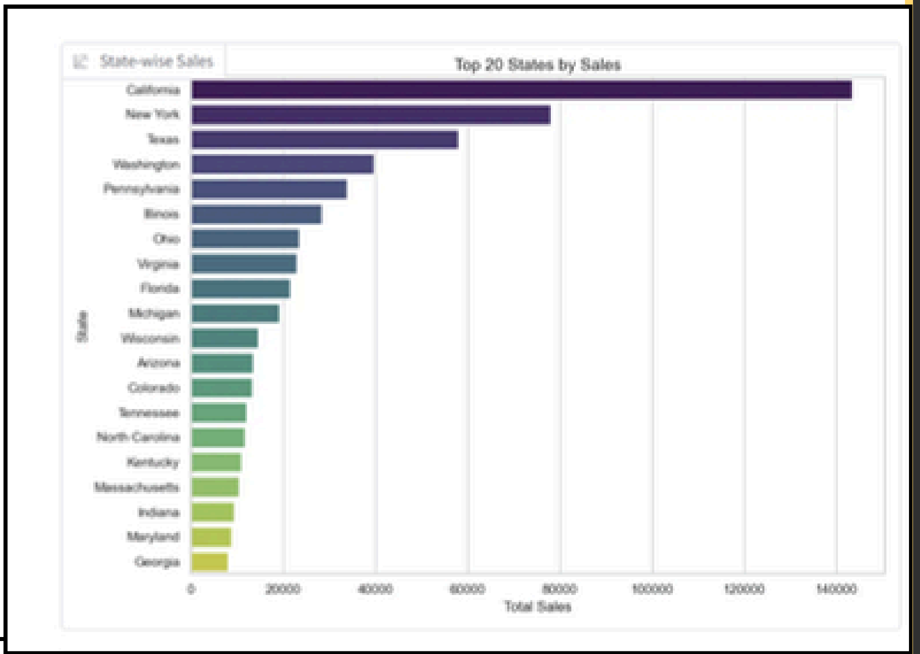
Module 6 – Price Recommendation

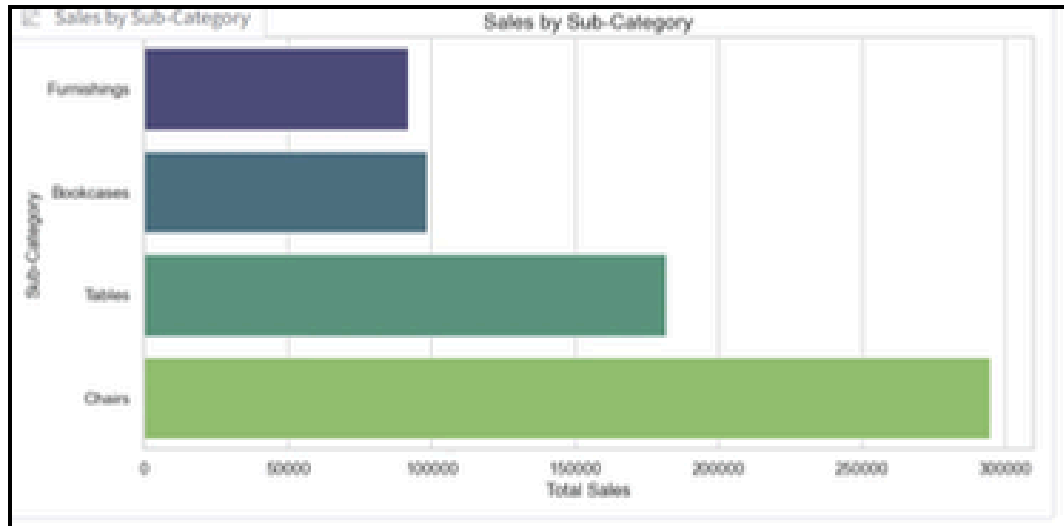
- Predicts product demand
- Suggests reorder quantity
- Helps prevent overstock & stockouts
- Uses historical sales + trend analysis

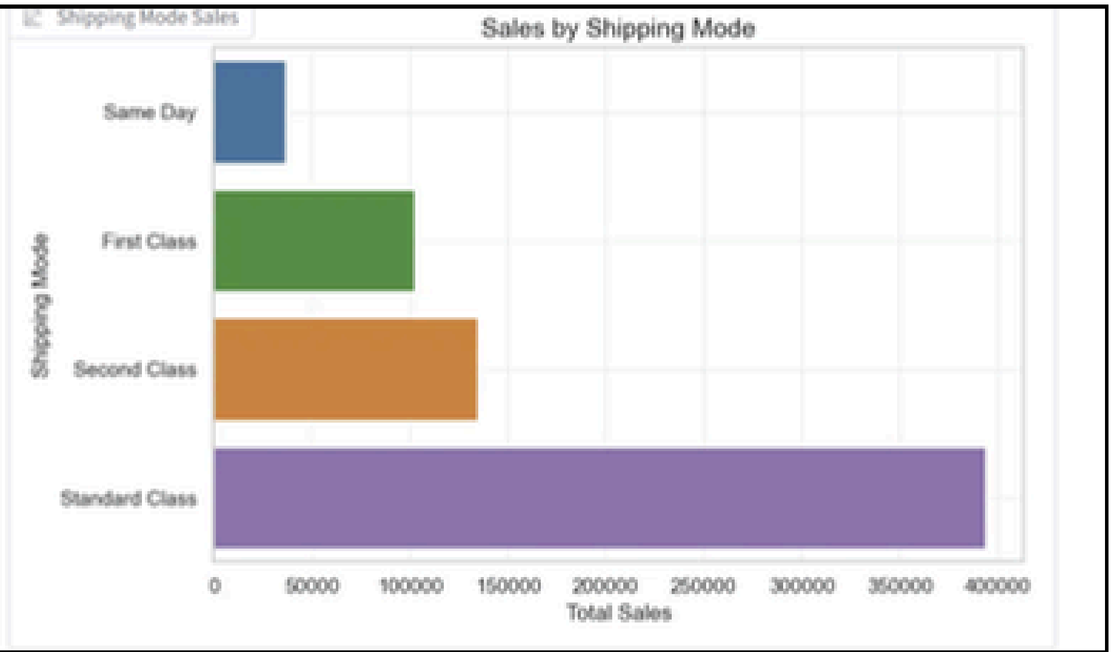
Analytics

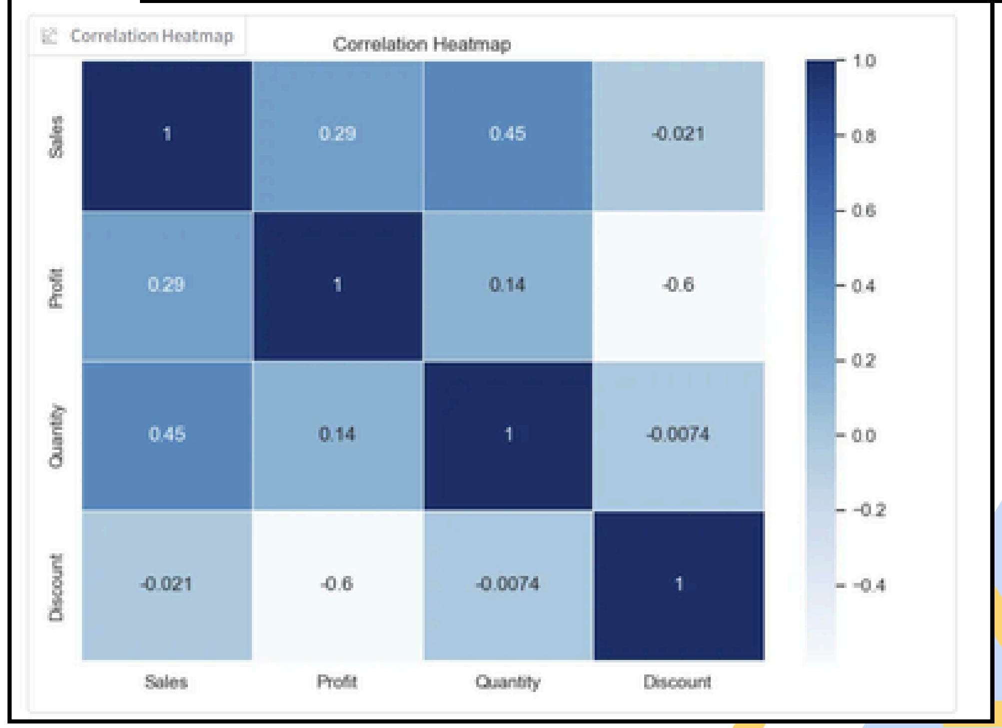

Monthly Sales Trend
Monthly Sales Trend
Total Sales
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Month


Yearly Sales Analysis
Yearly Sales Analysis
Total Sales
Year


State-wise Sales
Top 20 States by Sales
State
Total Sales

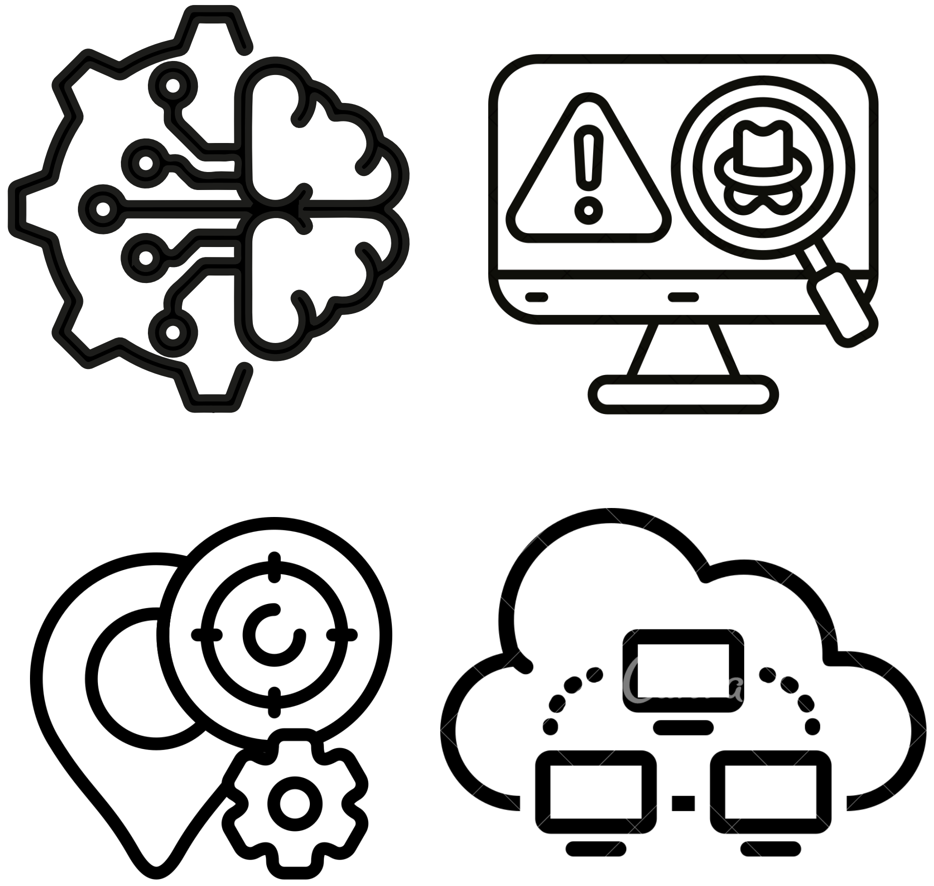

Sales by Sub-Category
Sales by Sub-Category
Furnishings
Bookcases
Tables
Chairs
Sub-Category
Total Sales


Shipping Mode Sales
Sales by Shipping Mode
Same Day
First Class
Second Class
Standard Class
Shipping Mode
Total Sales


Correlation Heatmap
Correlation Heatmap
Sales
Profit
Quantity
Discount

Future Scope

- Real-time database integration
- LSTM/Deep Learning forecasting
- Automated pricing optimization (AI-driven)
- Intelligent recommendation engine
- Fraud & anomaly detection
- Cloud-based scalable deployment
- Geo-Analytics visualizations



Conclusion

Our project successfully integrates key retail analytics modules into one decision-support system.

Sales forecasting provides accurate monthly predictions to guide planning, while the inventory module converts these forecasts into safety stock and reorder points.

Segmentation and association rules uncover customer patterns and product relationships that support targeted marketing and cross-selling.

The price recommendation engine offers data-driven pricing insights to boost profitability.

Overall, the solution strengthens retail decision-making by improving forecasting accuracy, inventory efficiency, customer understanding, and strategic pricing.

Reference

https://towardsdatascience.com/tagged/retail-analytics

https://hbr.org/topic/analytics

https://www.mckinsey.com/industries/retail/our-insights

ttps://journalofbigdata.springeropen.com/articles/10.1186/s40537-019-0265-5

Thank You